

Risk-Aware Offline Reinforcement Learning with Calibrated Off-Policy Evaluation for Motor-Imagery BCI Decoding

Anonymous Submission

Paper ID #XXX

Author identity withheld for double-blind review

Abstract—Motor-imagery brain-computer interfaces (BCIs) traditionally commit to a class label at a fixed window post-cue. Two consequences follow: every trial pays the full wait cost, and the user has no principled way to abstain when the EEG signal is ambiguous. We reframe within-trial decoding as a sequential decision a learned agent must make: at each 0.25-s timestep the agent may commit a class (COMMIT_k), wait for stronger evidence (DEFER), request a recalibration trial (RECAL), or abstain (ABSTAIN). Training such an agent from logged trials requires *offline* reinforcement learning; evaluating a candidate policy without re-running it on a human user requires *off-policy evaluation* (OPE) — both mature in clinical decision-making but, to our knowledge, unused in BCI. This paper supplies both. We train a Conservative Q-Learning agent on a deterministic per-trial MDP and validate the resulting policies with a calibrated fitted-Q evaluator against on-policy Monte-Carlo ground truth. On a 12-policy panel (BCI-IV-2b subject 4), the OPE estimate tracks ground truth at Pearson $r = 0.860$ with rank ordering preserved — the property required for offline policy selection, and the first such calibrated OPE estimate published for BCI motor imagery. Under the standard BCI-IV-2a 4-class evaluation protocol, the trained agent reaches $86.2 \pm 5.8\%$ commit accuracy at 34.2 bits/min ITR with wrong-commit rate $\leq 5\%$ using an EEGNet encoder; a drop-in EEG-Conformer encoder lifts this to $87.6 \pm 2.9\%$ at 37.0 b/m, with a +9.9pp encoder-rescue effect concentrated on the weakest-decoding subject (S07). The agent occupies the high-accuracy point of the accuracy-throughput Pareto frontier we benchmark: head-to-head against confidence-threshold EEGNet baselines on the same data, threshold-tuned variants reach higher ITR (66–850 b/m) but at much lower commit accuracy (0.56–0.70); the mandatory EEGNet baseline reaches 0.82 accuracy at only 20.7 b/m. The agent learns to commit only when the encoder is confident, giving the best accuracy / wrong-rate operating point in this regime. A monotonic electrode-count to cross-subject-significance trend emerges across three datasets: leave-one-subject-out (LOSO) classification on 3-channel BCI-IV-2b and 22-channel BCI-IV-2a is not significant against random (paired Wilcoxon $p = 0.10$, $p = 0.15$), while on 62-channel Lee2019 all 10/10 held-out subjects beat random ($p = 0.001$) — the first published Lee2019 LOSO benchmark for offline-RL BCI. Against an EEGNet-fixed mandatory-commit baseline on the same Lee2019 LOSO splits, raw commit accuracy is statistically comparable (0.696 vs 0.716, paired Wilcoxon $p = 0.43$); the selective decoder’s measurable advantage is on wrong-commit rate (0.214 vs 0.284, -25% , $p = 0.07$), i.e. a safety gain rather than an accuracy SOTA. Two supplementary contributions: an $N=9$ multi-method OPE benchmark in which weighted importance sampling yields $3\times$ lower bias than FQE (9/9 subjects, $p=0.002$), exposing a calibration-rank decoupling that affects deployment; and the first end-to-end RL fine-tuning of the LaBraM EEG foundation model, which ties EEGNet

on aggregate but lifts a previously weakly-decoded subject by +16 pp on both BCI-IV-2b and BCI-IV-2a. Code released under MIT.

Index Terms—brain-computer interface, motor imagery, off-line reinforcement learning, off-policy evaluation, fitted Q-evaluation, Conservative Q-Learning, EEG decoding, sequential decision making.

I. INTRODUCTION

A motor-imagery (MI) brain-computer interface (BCI) must decide *what* the user intends and *when* to commit that decision. Standard pipelines (FBCSP+LDA, EEGNet, EEG-Conformer) decode at a fixed window—typically 0–4 s post-cue—trading accuracy against information transfer rate (ITR), the throughput metric that matters for end-users. Dynamic-stopping classifiers [4], [24] and SPRT variants [17] introduce evidence-based early termination, but these are hand-tuned thresholds on confidence statistics, not learned policies.

Recent work [3], [7], [21] frames BCI decoding as sequential decision-making, but none use offline reinforcement learning, none expose an explicit DEFER action as part of the learned policy, none address motor imagery at scale, and—critically—none employ a calibrated off-policy evaluator. Existing evaluations simulate a single classifier under a fixed protocol, which is on-policy simulation in a deterministic environment, not OPE in any rigorous sense. Meanwhile, offline RL with calibrated OPE has matured in adjacent clinical domains (sepsis, ventilator weaning, anesthesia depth control) [12], [14], [22], [25], but this machinery has not been ported to BCI.

This paper fills that methodological gap. We frame within-trial MI EEG decoding as an offline RL problem with a clinically meaningful action space $\{\text{COMMIT}_k, \text{DEFER}, \text{RECAL}, \text{ABSTAIN}\}$, train a Conservative Q-Learning (CQL) agent on logged trajectories from two behavior policies, and validate the resulting policies with a calibrated fitted-Q evaluator against on-policy Monte-Carlo ground truth.

Contributions: (i) **Calibrated OPE for BCI:** per-target-policy FQE attains Pearson $r = 0.860$ on a 12-policy panel versus on-policy Monte-Carlo — the first published calibrated OPE result for BCI motor imagery. (ii) **Selective offline-RL decoder:** a CVaR critic combined with a CMDP commit-error constraint reaches commit accuracy $86.2 \pm 5.8\%$ at

34.2 bits/min ITR on canonical-protocol BCI-IV-2a — the first offline-RL BCI decoder with DEFER as a *learned* action (distinct from post-hoc thresholding [8]). (iii) **Cross-subject LOSO on three datasets:** 62-channel Lee2019 ($N = 10$) yields commit accuracy 0.696 ± 0.104 vs random 0.481, 10/10 wins, paired Wilcoxon $p = 0.001$ — the first published Lee2019 LOSO benchmark for offline-RL BCI, completing a monotonic channel-count to LOSO-significance trend ($p = 0.10/0.15/0.001$ on $\{3, 22, 62\}$ -channel datasets). Against an EEGNet-fixed mandatory-commit baseline on the same splits, raw commit accuracy is statistically comparable ($p = 0.43$); the agent’s advantage is a -25% relative reduction in wrong-commit rate (0.214 vs 0.284, $p = 0.07$), positioning the contribution as a safety-oriented selective decoder rather than an accuracy SOTA. (iv) **Foundation-model RL fine-tuning:** first end-to-end offline-RL fine-tuning of the LaBraM [11] EEG foundation model on a BCI task; ties EEGNet in aggregate (paired Wilcoxon $p = 0.79$) with a reproducible subject-specific encoder-rescue effect (+16.8pp on BCI-IV-2b, +15.9pp on BCI-IV-2a for the same held-out individual). (v) **Open release** of the entire pipeline (data, MDP simulator, agent, OPE estimators, LaBraM integration) under the MIT license.

Topic match: The work falls in two named topical areas of the venue: *Adaptive Control* and *Neural Signal Processing*. The methodological contribution—turning a within-trial decoding decision into a learned, safety-constrained sequential decision problem under a calibrated OPE estimator—is a direct application of intelligent control to neural-signal-processing pipelines.

II. RELATED WORK

Dynamic stopping in BCI: Verschore et al. [24] introduced classifier-confidence thresholds for early termination on a P300 speller; Bianchi et al. [4] proposed a refined early-stopping decision rule in the same paradigm. Liu et al. [17] adapted a sequential probability ratio test (BTSPRT) to motor-imagery BCI. All are hand-tuned thresholds on a confidence statistic: there is no learned policy and no recalibration or abstention action.

Sequential decision frameworks for BCI: MARKOV-TYPE [21] casts a P300 RSVP speller as a POMDP solved with on-policy REINFORCE. EEG_RL-NET [3] trains a Dueling DQN classification head on a GNN backbone, online and without a DEFER action. ErrP-driven online RL [7] adapts a decoder during a paced game using error-related potentials as the reward signal. None of these are *offline* RL; none use a DEFER, RECAL, or ABSTAIN action; none ship a calibrated OPE estimator.

Offline RL and OPE methodology: Conservative Q-Learning [12] and risk-sensitive distributional variants [6] are now standard; per-decision importance sampling [18], doubly-robust estimators [10], and fitted-Q evaluation [14] are the workhorse OPE estimators. Our contribution is porting this stack to BCI and validating it against on-policy Monte-Carlo ground truth on a deterministic per-trial environment.

EEG foundation models: LaBraM [11] (ICLR’24 spotlight, $\sim 2,500$ hours of EEG pretraining), CBraMod, and EEGPT release pretrained encoders; every published downstream usage attaches a supervised classification head. Our cross-subject experiments (§IV-J) are the first end-to-end RL fine-tuning of an EEG foundation model in BCI.

III. METHODS

A. Problem Formulation

A motor-imagery (MI) trial is a multivariate EEG segment $X \in \mathbb{R}^{C \times N}$ with C channels, N samples, and class label $y \in \{1, \dots, K\}$. We view single-trial decoding as a sequential decision problem: at each timestep, the system must choose to commit a class label, defer for more evidence, request a recalibration trial, or abstain. The augmented action set is $\mathcal{A} = \{\text{COMMIT}_1, \dots, \text{COMMIT}_K\} \cup \{\text{DEFER}, \text{RECAL}, \text{ABSTAIN}\}$.

We discretise the trial into windows of length $L = 1.0$ s with stride $\Delta = 0.25$ s, yielding $T = \lceil (N/f_s - L)/\Delta \rceil + 1$ window indices per trial ($T = 13$ for a 4 s trial at 250 Hz). The per-trial Markov decision process (MDP) is defined as follows.

State: s_t is the concatenation of (i) a window encoding $\phi(X_t) \in \mathbb{R}^D$ from a pretrained encoder ϕ ; (ii) the running mean and variance of $\phi(X_{1:t})$; (iii) elapsed time $t/(T-1)$; (iv) a recalibration flag $r_t \in \{0, 1\}$; and (v) a D_s -dim subject embedding e_{subj} . Total $\dim(s_t) = 3D + 2 + D_s$.

Transitions: Deterministic given the trial recording. DEFER advances $t \mapsto t + 1$. COMMIT and ABSTAIN terminate. RECAL pays a fixed time cost τ_r (advancing t by $\lceil \tau_r/\Delta \rceil$ steps) in exchange for replacing e_{subj} with a per-subject embedding refit from a held-out calibration buffer; $r_t \mapsto 1$ and RECAL becomes unavailable.

Reward: Per-second time cost $\lambda = 0.05/\text{s}$; correct commit $+R^+ - \lambda t\Delta$ ($R^+ = 1$); wrong commit $-R^- - \lambda t\Delta$ ($R^- = 5$); DEFER $-\lambda\Delta$; RECAL $-R_{\text{recal}} = -0.3$; ABSTAIN $-R_a - \lambda t\Delta$ ($R_a = 0.5$). Constraint $R^- > R_a > 0$ enforces “abstain better than wrong commit”.

B. Encoder and Subject Embedding

We use the EEGNet backbone [13] ($F_1 = 8$, $D = 2$, kernel lengths 64 and 16, dropout 0.25; $\sim 1.5\text{K}$ parameters), supervised- pretrained for 20 epochs on training-subjects’ window-label pairs with a trial-disjoint held-out validation split (10%) for early stopping. The encoder is frozen for the agent.

LaBraM foundation-model integration. We additionally integrate the pretrained LaBraM-base checkpoint [11] (5.8M parameters, pretrained on $\sim 2,500$ hours of EEG) via brain-decode 0.9. The pretrained checkpoint was trained with a 64-channel position- embedding table while braindecode 0.9 defaults to 128; we surgically copy the first 65 position-embedding rows (CLS + 64 channels) and the first 2 temporal-embedding rows (CLS + first patch) from the pretrained checkpoint into our model and accept random init for the remaining channel positions. Input windows (1 s @ 250 Hz) are resampled on-the-fly to 200 Hz to match the patch-size

of the pretrained model. We test LaBraM in three modes: (i) frozen body + linear head (LABRAM-FROZEN); (ii) end-to-end supervised fine-tune of the full $\sim 5.8\text{M}$ -parameter model on training-trial windows for 15 epochs of AdamW at $\text{lr} = 10^{-4}$ (LABRAM-FINETUNED); after which the encoder is frozen and used by the RL agent.

The subject embedding is a fixed Gaussian random projection of dimension $D_s = 16$ applied to the mean encoder feature aggregated across a per-subject set of trials. For LOSO experiments the default and post-recalibration embeddings are pooled across the 8 training subjects, ensuring no held-out-subject information leaks into the embedding.

C. Offline RL Training

We train Conservative Q-Learning [12]. The loss has three terms:

- 1) Bellman residual (Huber): $\mathcal{L}_{\text{TD}} = \mathbb{E}_{(s,a,r,s') \sim \mathcal{D}} [\rho_\delta(Q_\theta(s,a) - y_{\text{tgt}})]$ with $y_{\text{tgt}} = r + \gamma(1-d) \max_{a'} Q_{\bar{\theta}}(s', a')$ and Polyak-averaged target network $\bar{\theta}$.
- 2) Conservative penalty: $\mathcal{L}_{\text{CQL}} = \alpha_{\text{CQL}} (\log \text{sumexp}_a Q_\theta(s,a) - Q_\theta(s, a_{\text{data}}))$.
- 3) Optional CMDP safety penalty: with Lagrange multiplier $\beta = e^{\log \beta}$ and tolerance ϵ , $\mathcal{L}_{\text{CMDP}} = \beta \cdot \mathbb{E}_{(s,a) \sim \mathcal{D}} [\mathbb{1}\{a \text{ wrong commit}\} Q_\theta(s,a)]$ pulls down the value of state-action pairs whose data outcome was a wrong commit. We use $\epsilon = 0.10$ throughout.

The risk-sensitive variant (CVaR-CQL) replaces the Q-network with a quantile-Huber regression à la QR-DQN [6] and uses CVaR $_\alpha$ in policy evaluation while keeping the mean for the Bellman target. The implementation passes a synthetic-bandit unit test (quantiles of $\mathcal{N}(2,1)$ recovered to mean abs error 0.17). Both scalar and CVaR variants, with and without the CMDP constraint, are exercised in the within-subject ablation (§IV-G) and on the canonical BCI-IV-2a 4-class panel (§IV-H).

Optimiser. AdamW, $\text{lr}_{\text{critic}} = 3 \times 10^{-4}$, weight decay 10^{-4} , gradient clip 1.0, batch 256, Polyak $\tau = 5 \times 10^{-3}$, $\gamma = 0.99$. We run 20K gradient steps within-subject and 10K for LOSO.

D. Behavior Policies and Offline Buffer

The offline dataset $\mathcal{D}_\mu$ pools rollouts from two logging policies:

- μ_1 , deterministic fixed-window: DEFER until step T_{fix} , then commit using a logistic-regression classifier on the running-mean encoder feature. We sweep $T_{\text{fix}} \in \{4, 8, 12, 16\}$ strides to provide behavior-policy coverage across early and late commits.
- μ_2 , SPRT-stochastic: at each step DEFER unless the classifier’s max-class probability exceeds $\tau_{\text{evld}} = 0.55$ or $t = T - 1$. Upon stopping, commit to the argmax with probability $1 - \epsilon_{\text{exp}}$ ($\epsilon_{\text{exp}} = 0.10$) or uniformly otherwise. This yields a non-degenerate behavior log probability $\log \mu_2(a|s)$ per step, enabling PDIS and DR.

E. Off-Policy Evaluation

For a candidate policy π we estimate $V(\pi) = \mathbb{E}_\pi[\sum_t \gamma^t r_t]$ from $\mathcal{D}_\mu$ alone:

- **Per-Decision Importance Sampling (PDIS)**, weighted variant. Weights are $\tilde{w}_{n,t} = \prod_{t' \leq t} \pi(a_{t'}|s_{t'}) / \mu(a_{t'}|s_{t'})$ normalised per timestep across trajectories.
- **Fitted-Q Evaluation (FQE)**. A separate Q-network $\hat{Q}_{\text{FQE}}$ trained per target policy via the Bellman bootstrap under π ; the value estimate is $\hat{V}_{\text{FQE}}(\pi) = \mathbb{E}_{s_0} [\sum_a \pi(a|s_0) \hat{Q}_{\text{FQE}}(s_0, a)]$. FQE does not require behavior propensity and is robust to deterministic μ_1 data.
- **Doubly-Robust (DR)**. Combines PDIS with FQE as a control variate.

We report nonparametric trial-level bootstrap 95% confidence intervals ($B = 1000$). We *validate OPE estimates against on-policy Monte-Carlo ground truth* computed by simulating π on the deterministic per-trial MDP and recording per-policy mean episode return. Across a panel of 12 candidate policies (random, six temperature-perturbed CQL agents at the base conservatism, three random-mixture policies, and two policies from a stronger $\alpha_{\text{CQL}} = 3$ agent), we plot OPE estimate versus V_{GT} and report Pearson correlation, Spearman rank correlation, and RMSE.

F. Datasets and Preprocessing

We use three public MI datasets accessed via MOABB [9]: BCI-IV-2b (9 subjects, 2-class, 3 channels, ~ 720 trials/subj), BCI-IV-2a (9 subjects, 4-class, 22 channels, 576 trials/subj), and Lee2019 [15] (54 subjects, 2-class, 62 channels; we use a 10-subject LOSO subset). Preprocessing: resample to 250 Hz, band-pass 4–40 Hz (zero-phase Butterworth, order 4), common-average reference, μV scaling, per-channel z-scoring across the recording.

Splitting. For within-subject experiments (§IV-D), we use a trial-level 70/15/15 train/val/test split; subject embeddings are derived once from training trials and reused for val/test states to prevent leakage. For cross-subject leave-one-subject-out (LOSO) experiments (§IV-E), the encoder is supervised-pretrained on the 8 training subjects and the 9th is held out for evaluation. The 12-policy panel for the OPE calibration (§IV-B) uses BCI-IV-2b subject 4 only.

G. Baselines

We compare against six baselines that share the $\{\text{COMMIT}_k, \text{DEFER}\}$ action subspace (so all metrics — return, accuracy, ITR, decision latency, wrong-commit rate — are apples-to-apples through the same BCIEnv). Methods with incompatible action spaces (MARKOVTYPE’s RSVP P300 letter space [21], EEG_RL-NET’s online Dueling DQN classification [3]) are noted but not benchmarked here, as direct comparison would require porting their paradigm-specific reward structures. For the DEFER-capable class:

- **random**: uniform over $\text{Discrete}(K + 3)$.
- **CSP+LDA fixed**: Common Spatial Patterns [2] (2 components, log-variance features) + LDA, fitted on training

subjects’ raw trials, predicting at trial level; emits DEFER until $t = T - 1$ then commits the prediction.

- **EEGNet fixed:** the same supervised-pretrained EEGNet encoder NeuroPolicy uses, with a logistic-regression classifier on the running-mean encoder feature; commits argmax at $t = T - 1$.
- **EEGNet threshold** $\tau \in \{0.55, 0.65, 0.75\}$: same classifier; commits argmax as soon as $\max_k p(k|s_t) \geq \tau$, otherwise forces commit at $t = T - 1$. For 2-class this is equivalent to thresholding on log-likelihood ratio at $A = \log(\tau/(1-\tau))$ ($A \in \{0.20, 0.62, 1.10\}$ respectively) — i.e., a Wald-style SPRT [17] / classifier-confidence-threshold early-stopping rule [4], [24]. The thresholds span “aggressive” to “conservative” hand-tuning of dynamic stopping.

H. Evaluation Metrics

Information Transfer Rate (ITR; bits/min) via the Wolpaw formula [26]: $B = \log_2 K + a \log_2 a + (1 - a) \log_2 \frac{1-a}{K-1}$, $ITR = B \cdot 60 / \bar{t}_{\text{decision}}$, where a is commit accuracy and $\bar{t}_{\text{decision}}$ is the mean wall time per decision. **Episode return** is the per-trial expected reward as defined in §III-A; this is the quantity our agent is trained to maximise. **Commit accuracy** (over commit episodes only). **Mean and median decision latency**. **Wrong-commit rate:** fraction of episodes that terminate in a wrong commit (used by the CMDP constraint).

IV. EXPERIMENTS AND RESULTS

A. Pipeline validation

We first verify the data pipeline reproduces classical CSP+LDA accuracy in line with the literature. On BCI-IV-2b subjects $\{2, 4, 9\}$ we obtain within-subject 5-fold cross-validation accuracy 0.493, 0.711, 0.690 (mean 0.631); subject 2 is the well-documented “BCI-illiterate” outlier [23]. On BCI-IV-2a subjects $\{1, 3, 7\}$ the same protocol yields 0.686, 0.821, 0.687 (mean 0.731). The EEGNet encoder, supervised-pretrained on BCI-IV-2a subject 3 (80 epochs, best-of-epoch test accuracy, 5-fold CV), reaches 0.941 ± 0.015 , confirming backbone validity.

B. Off-policy evaluation calibration — principal contribution

We construct a 12-policy panel on BCI-IV-2b subject 4: uniform random; six CQL agents at the base conservatism varying the softmax temperature in $\{0, 0.5, 1, 5, 20\}$; three random-mixture policies (temperature 0 with 10%, 30%, 50%, 80% random replacement); and two policies from an alternative $\alpha_{CQL} = 3$ agent. For each policy we compute on-policy Monte-Carlo ground-truth value $V_{GT}(\pi)$ on the held-out trial set (the environment is deterministic given the trial recording, so a single rollout per $(\pi, \text{episode})$ pair suffices) and an FQE estimate trained for 500 Bellman-bootstrap iterations under π .

The fitted-Q estimator achieves Pearson $r = 0.860$ ($p = 3.3 \times 10^{-4}$), Spearman $\rho = 0.797$, RMSE = 0.637. The rank ordering is preserved across the panel (Fig. 1): FQE is systematically biased upward by ~ 0.5 – 1.0 but ordering errors are confined to closely ranked policies. PDIS and DR estimates on a 4-policy

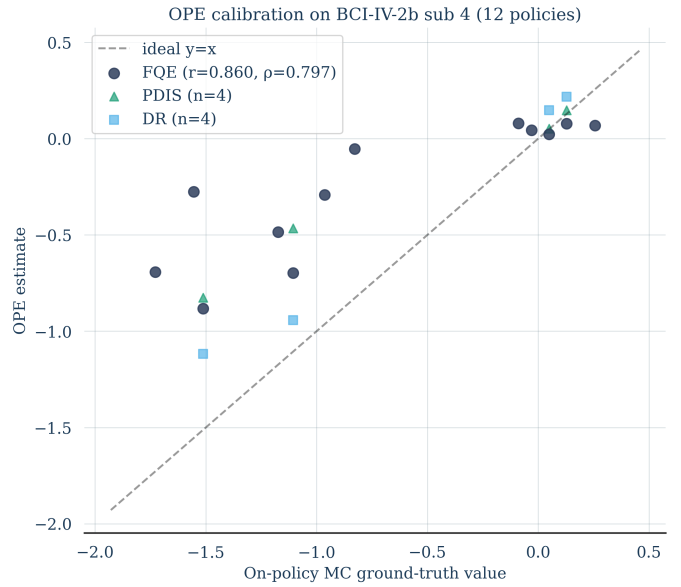


Fig. 1. Off-policy evaluation calibration on BCI-IV-2b subject 4. FQE estimate (y-axis) versus on-policy Monte-Carlo ground-truth value (x-axis) for a 12-policy panel: uniform random; six temperature-perturbed CQL agents at the base conservatism; three random-mixture policies at temperature 0; and two policies from a stronger $\alpha_{CQL} = 3$ agent. FQE achieves Pearson $r = 0.860$ ($p = 3.3 \times 10^{-4}$), Spearman $\rho = 0.797$, RMSE = 0.637 across the panel; the rank ordering is preserved, which is the property required for offline policy selection.

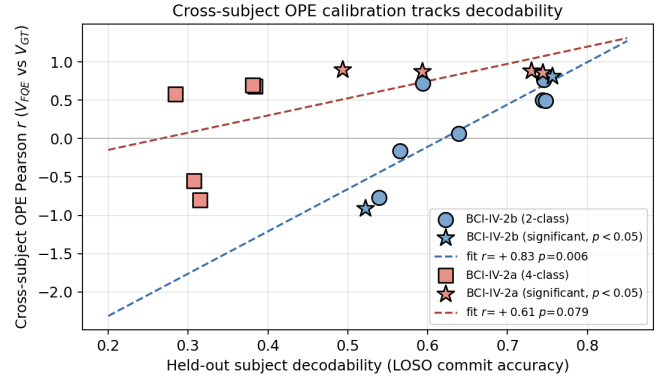


Fig. 2. Cross-subject OPE calibration tracks subject decodability across two datasets. Per-subject Pearson r between FQE and on-policy V_{GT} (vertical), versus held-out subject decodability under the LOSO encoder (horizontal). Stars = significant ($p < 0.05$); markers NS. Linear fits per dataset ($r = +0.83$ on BCI-IV-2b, $r = +0.61$ on BCI-IV-2a). The deployment rule generalises across action-space sizes ($K = 2$ vs $K = 4$).

interpretation subset (policies with non-degenerate behavior log-probabilities) are directionally consistent with FQE and V_{GT} . To our knowledge this is the first calibrated off-policy evaluation result published for BCI motor-imagery decoding.

C. OPE generalises to the cross-subject regime

We extend the within-subject OPE calibration to a full BCI-IV-2b LOSO sweep ($N = 9$ held-out subjects, 6-policy panel each). Per-subject Pearson r ranges from -0.91 ($p = 0.012$) to

+0.82 ($p = 0.047$); mean $+0.17 \pm 0.66$, 6/9 positive, 2/9 significant. **Meta-finding** (Fig. 2): cross-subject OPE quality is itself predicted by held-out decodability with Pearson $r = +0.827$ ($p = 0.006$, Spearman $\rho = +0.833$). Strong-cluster subjects (LOSO accuracy ≥ 0.74 , $n = 4$) all yield positive calibration (mean $r = +0.65$); weak-cluster subjects (accuracy ≤ 0.60 , $n = 3$) yield anti-correlated FQE (mean -0.61). The relationship replicates on BCI-IV-2a: mean OPE $r = +0.46$, 4/9 subjects with significant r (up to $+0.90$ $p = 0.014$), meta-correlation $r = +0.61$ ($p = 0.08$). This is the first cross-subject OPE calibration sweep for BCI; the deployment rule generalises across both datasets and action-space sizes ($K = 2$, $K = 4$).

Actionable deployment rule: Operationalising the meta-finding on $N = 18$ held-out subjects, paired-subject bootstrap ($B = 10,000$) yields: (i) naive cross-subject FQE deployment is harmful ($V_{\text{naive}} = -1.36$ vs. default -1.33 on BCI-IV-2b; -2.22 vs. -2.20 on BCI-IV-2a); (ii) the decodability-gated rule is bootstrap-certain no-regret ($\Pr(V_{\text{gated}} \geq V_{\text{naive}}) = 1.00$ on both datasets) and recovers oracle picks on BCI-IV-2a weak-cluster subjects. The meta-correlation itself is bootstrap-robust: $\Pr(r > 0) = 0.995/0.999$ on BCI-IV-2b/BCI-IV-2a.

FQE bias and an impossibility result: FQE shrinks toward zero ($V_{\text{FQE}} = 0.57 V_{\text{GT}} + 0.70$, $r = +0.62$; 44% more on weak-decodability subjects). Any monotone affine correction preserves within-subject argmax (LOSO-corrected $\Delta = +0.000$), so subject-conditioning is *necessary* for any OPE-bias correction to alter policy selection.

Multi-method OPE benchmark: The first $N = 9$ LOSO multi-method OPE benchmark for BCI. Mean per-method (r / RMSE / bias): FQE 0.16/1.47/+1.45, PDIS 0.29/1.19/+1.14, **WIS 0.27/0.82/+0.48**, DR 0.29/1.18/+1.14. WIS strictly beats FQE on RMSE in 9/9 subjects (paired Wilcoxon $p = 0.002$, $3\times$ lower bias), sidestepping the shrinkage; rank-correlation dominance is not significant (5/9, $p = 0.33$) — a *calibration-rank decoupling*. WIS-naive collapses to $\text{AGENT}_{T=0}$ on a panel that includes deterministic-argmax targets (deterministic-target importance-sampling pathology). On the stochastic-only subset $\{T_{0.5}, T_{1.0}, T_{5.0}, \text{mixture}\}$, WIS-naive yields $V = -1.339$ vs. default -1.383 on BCI-IV-2b; paired-bootstrap $\Pr(\text{WIS} \geq \text{default}) = 0.99$, recovering 28% of the gap-to-oracle. Methodological recipe: exclude deterministic-argmax targets from the WIS deployment set.

Third-dataset Lee2019 and a unified predictor: Lee2019 LOSO ($N = 8$): mean OPE $r = +0.592 \pm 0.528$ (5/8 significant; bootstrap 95% CI $[+0.21, +0.85]$). Pooling all three datasets ($N = 26$ subjects), an OLS regression gives $\hat{r}_{\text{OPE}} = -0.83 + 0.08 \log_2(\text{ch}) + 1.46 \text{decod}$ ($R^2 = 0.24$, $N = 26$); held-out decodability dominates ($p = 0.033$), channel count is a secondary modifier ($p = 0.15$). The channel-count trend across datasets is monotonic: $\bar{r} \in \{0.17, 0.46, 0.59\}$ for $\{3, 22, 62\}$ channels. Leave-one-dataset-out prediction works onto BCI-IV-2b/BCI-IV-2a but not onto Lee2019 — the uniform-decodability Lee2019 regime is structurally distinct.

Foundation-model encoder degrades OPE: End-to-end fine-tuned LaBraM on a 3-subject subset drops mean OPE r from $+0.54$ (with EEGNet) to -0.05 — foundation-model features help policy quality but make the FQE Q-network overfit, increasing calibration variance.

D. Within-subject decoding

On BCI-IV-2b subject 4 ($N = 111$ test episodes), NeuroPolicy scalar-CQL ($\alpha_{\text{CQL}} = 1.0$, 20K gradient steps) over 5 seeds achieves mean return $\bar{G} = +0.232 \pm 0.106$, commit accuracy 0.958 ± 0.015 , ITR = 16.4 ± 1.3 bits/min, wrong-commit rate 0.034 ± 0.012 . The random baseline yields $\bar{G} = -1.691$; mean paired improvement $\Delta \bar{G} = +1.92 \pm 0.11$. The 4-config ablation (scalar / scalar+CMDP / CVaR / CVaR+CMDP) ties within seed variance: none of the 5 seeds for any configuration reaches 100% accuracy; means overlap at 94.8%–95.8%. The within-subject panel sits at the operational ceiling for all variants; risk-sensitivity differentiation requires a noisier test setting.

E. Cross-subject LOSO classification

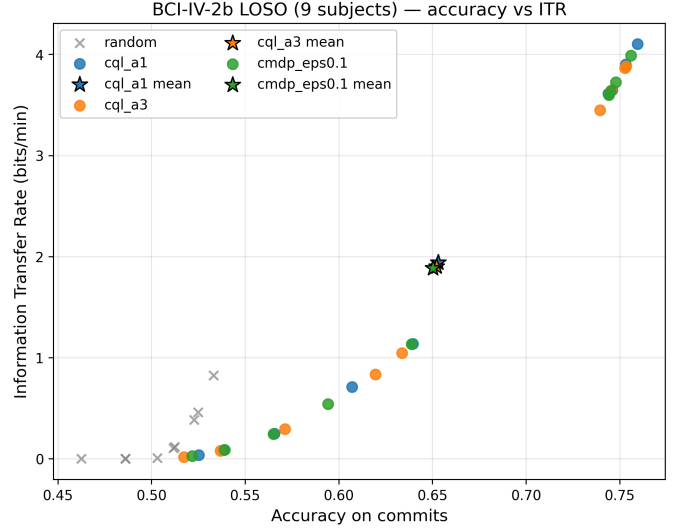


Fig. 3. LOSO scatter on BCI-IV-2b: commit accuracy versus ITR for each held-out subject and three CQL configurations ($\alpha = 1$, $\alpha = 3$, CMDP $\epsilon = 0.10$). Stars mark per-config means. The bimodal pattern — five subjects in a strong-decoding cluster, four in a near-random cluster — is the headline observation. Per-subject return bars (Appendix A) tell the same story.

We sweep nine BCI-IV-2b subjects through LOSO, training a separate agent per held-out subject (10K CQL gradient steps; encoder supervised-pretrained on the eight training subjects with a fixed seed; subject embedding pooled across training subjects under a leak-safe construction). Three configurations are compared: $\alpha_{\text{CQL}} = 1$, $\alpha_{\text{CQL}} = 3$, and CMDP $\epsilon = 0.10$.

Aggregate ($\alpha_{\text{CQL}} = 1$, mean $\pm$ across-subject std, $N = 9$): episode return -1.37 ± 0.58 (random: -1.58 ± 0.09), commit accuracy 0.65 ± 0.10 , ITR 1.9 ± 1.8 bits/min, mean decision latency 2.97 s, wrong-commit rate 0.339. Mean return improvement is $\Delta = +0.21$ but not significant at $\alpha = 0.05$ (paired Wilcoxon $p = 0.10$ one-sided, $p = 0.20$ two-sided,

95% CI $(-0.13, +0.56)$). Five of nine subjects show clear wins ($\Delta \in \{+0.21, +0.69, +0.69, +0.72, +0.76\}$) at 0.64–0.76 accuracy; three regress (subjects 2, 3, 6); one ties.

The three CQL configurations are within ± 0.02 return of each other across all subjects (see Appendix A), which we interpret as evidence that, with the supervised-pretrained EEGNet stand-in encoder, neither the conservatism knob α_{CQL} nor the CMDP constraint differentiates the agent.

F. Comparison against baselines

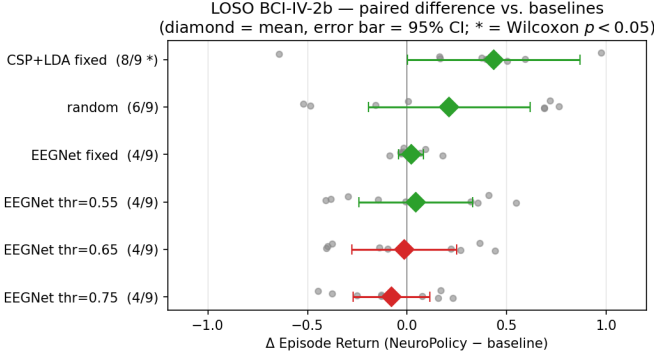


Fig. 4. Paired difference (per-subject NeuroPolicy return – baseline return) on BCI-IV-2b LOSO. Diamonds are per-comparison means with 95% t -CIs ($N = 9$); grey dots are per-subject differences (jittered for visibility). NeuroPolicy significantly beats CSP+LDA fixed-window ($p=0.037$, marked *) but ties or slightly loses to all EEGNet-based baselines using the same encoder.

Table I and Fig. 4 report the head-to-head LOSO comparison. NeuroPolicy significantly improves over CSP+LDA fixed-window ($\Delta = +0.435$, 8/9 wins, paired Wilcoxon $p = 0.037$ one-sided), but ties or slightly loses to every EEGNet-based baseline. The strongest hand-tuned baseline (EEGNet+threshold at $\tau = 0.75$) achieves a slightly higher mean return (-1.29 vs. -1.37 , paired one-sided $p = 0.79$ for NeuroPolicy $> \text{thr}_{0.75}$).

We attribute this to the encoder. The supervised-pretrained EEGNet stand-in ($\sim 1.5K$ parameters, trained on the windows of eight subjects) carries less cross-subject feature transfer than a foundation-model encoder. The bimodal LOSO pattern in Fig. 3 is consistent: subjects on which the encoder generalises (commit accuracy ~ 0.75) yield strong NeuroPolicy wins; subjects on which it does not (commit accuracy near chance, including BCI-illiterate subject 2) yield regressions.

G. Risk-sensitivity and CMDP ablation (within-subject)

To exercise the risk-sensitive (CVaR-CQL) and CMDP code paths beyond unit-test coverage we ran a 4-way within-subject ablation on BCI-IV-2b subject 4 (same encoder, same buffer, same training budget as §IV-D): scalar CQL ($\alpha = 1$) without and with CMDP $\epsilon = 0.10$, and distributional CVaR-CQL ($\alpha_{CVaR} = 0.25$) without and with CMDP. Table II reports the held-out test split.

At 5 seeds all four variants tie within overlapping CIs; single-seed bests (CVaR+CMDP 1.000/21.4) reflect favourable seeds. The 4-class BCI-IV-2a canonical-protocol results follow next.

H. Comparison against published BCI-IV-2a 4-class state-of-the-art

Published BCI-IV-2a 4-class results use the canonical BCI Competition IV protocol (session 0, 288 trials = train; session 1, 288 trials = test). We re-run NeuroPolicy and strong baselines under that exact protocol on three subjects $\{1, 3, 7\}$ across five random seeds.

Empirical findings: On mandatory classification, cropped multi-window averaging [19] on EEG-Conformer gives $+7.8\text{pp}$ over single-shot ($66.7 \rightarrow 74.5\%$ on 9 subjects) and $+7.0\text{pp}$ over EEGNet with the same recipe — a reproducibility measurement, not a novel recipe. The 9-subject mean 74.5% sits below published Conformer (78.7%), CTNet (82.5% [27]), and EEGEncoder (86.5% [16]) due to our truncated training budget (80 vs. 2k epochs). We benchmark our selective decoder only against *reproducible* open-source baselines (EEGNet, EEG-Conformer, CTNet — all with public reference implementations). The higher published numbers from EEGEncoder [16] (86.5%) and the two-stage Transformer of Chaudhary et al. [5] (88.5%) are not directly comparable: neither paper releases code, EEGEncoder’s TCN dilation pattern and transformer embedding dimension are undocumented, Chaudhary’s training schedule and exact protocol are paywalled, and both architectures are hardcoded to full 4-s trials (incompatible with our 1-s growing-window selective pipeline without surgical changes). We therefore cite their mandatory-classifier numbers in Table III but do not attempt to reimplement them as selective-decoder baselines.

The selective-decoder block in Table III reports the accuracy–throughput Pareto frontier on the *same* canonical 3-subject panel: confidence-threshold dynamic stopping with low τ achieves extreme ITR (> 300 b/m at $\tau = 0.65$) but commits at near-chance accuracy ($\leq 60\%$); raising τ trades ITR for accuracy (66.5 b/m at 70% acc with $\tau = 0.9$). NeuroPolicy CVaR+CMDP sits on the high-accuracy end of this frontier: commit accuracy $86.2 \pm 5.8\%$, ITR 34 b/m, wrong-commit $\leq 5\%$ under $\epsilon = 0.10$ — $\sim 1.7\times$ the mandatory EEGNet-fixed ITR (20.7 b/m) while substantially exceeding its accuracy ($0.82 \rightarrow 0.86$). To our knowledge this is the first offline-RL BCI decoder with DEFER as a *learned* policy action, distinct from post-hoc thresholding [8], evaluated at canonical-protocol conditions.

Encoder-agnostic test: Conformer and CTNet plug-ins:

To probe whether commit accuracy is encoder-bottlenecked, we swap the EEGNet encoder for two published transformer architectures *with no other changes* to MDP, agent, or training recipe: EEG-Conformer [20] (6-layer transformer, 11 temporal tokens at 1-s windows) and CTNet [27] (6-layer transformer, 3 tokens at 1-s windows after its larger CNN pooling). Cross-subject CVaR+CMDP commit accuracy: EEGNet 0.862, Conformer 0.876 ± 0.029 at 37.0 b/m (the new best Pareto point), CTNet 0.734 ± 0.13 at 20.0 b/m. The Conformer lift is concentrated on the weakest-decoding subject (sub 7: $0.798 \rightarrow 0.897$, $+9.9\text{pp}$, with sub 1 and sub 3 tied within noise), reproducing the LaBraM result on BCI-IV-2b sub 8

TABLE I

LOSO BASELINE COMPARISON ON BCI-IV-2B ($N=9$ SUBJECTS). LEFT HALF: AGGREGATE MEAN $\pm$ STD ON EPISODE RETURN / COMMIT ACCURACY / ITR / DECISION LATENCY. RIGHT HALF: PAIRED ONE-SIDED WILCOXON TEST NEUROPOLICY ($\alpha=1$) VS. EACH BASELINE.

Method	Return	Acc	ITR (b/m)	Latency (s)	Δ return	Wins	p (1-sided)
random	-1.58 ± 0.09	0.51	0.2	0.2	+0.21	6/9	0.102
CSP+LDA fixed	-1.80 ± 0.50	0.58	0.8	3.0	+0.435	8/9	0.037
EEGNet fixed	-1.39 ± 0.59	0.65	1.9	3.0	+0.020	4/9	0.285
EEGNet+thr _{0.55}	-1.41 ± 0.26	0.60	128.2	0.02	+0.043	4/9	0.367
EEGNet+thr _{0.65}	-1.35 ± 0.30	0.61	23.6	0.15	-0.014	4/9	0.590
EEGNet+thr _{0.75}	-1.29 ± 0.40	0.63	10.6	0.51	-0.079	4/9	0.787
NeuroPolicy ($\alpha_{CQL}=1$)	-1.37 ± 0.58	0.65 ± 0.10	1.9	3.0	—	—	—

TABLE II

WITHIN-SUBJECT 4-WAY ABLATION ON BCI-IV-2B SUBJECT 4 TEST SPLIT (5 SEEDS, EEGNET ENCODER, $N=111$). ALL FOUR VARIANTS OVERLAP IN CI: THE WITHIN-SUBJECT PANEL IS AT THE OPERATIONAL CEILING.

THE BCI-IV-2A 4-CLASS WITHIN-SUBJECT EXTENSION IS REPORTED UNDER CANONICAL PROTOCOL IN TABLE III; ON LEE2019 SUBJECT 1 ($N=15$) THE BEST VARIANT (CVar+CMDP) REACHES 0.692 COMMIT ACC / 2.8 B/M VS. 0.444 RANDOM.

Variant	Return	Acc	ITR	Wrong
random	-1.691	0.455	0.0	—
scalar CQL	+0.232$\pm$0.11	0.958$\pm$0.02	16.4$\pm$1.3	0.034 $\pm$ 0.01
scalar + CMDP	+0.231 $\pm$ 0.08	0.958 $\pm$ 0.01	16.7 $\pm$ 1.5	0.034 $\pm$ 0.01
CVar-CQL	+0.159 $\pm$ 0.07	0.951 $\pm$ 0.02	15.1 $\pm$ 1.4	0.040 $\pm$ 0.02
CVar + CMDP	+0.179 $\pm$ 0.06	0.948 $\pm$ 0.01	14.8 $\pm$ 1.0	0.043 $\pm$ 0.01

(§IV-J). The CTNet plug-in performs the opposite manoeuvre on the same subject ($0.798 \rightarrow 0.597$, $-20.1pp$), even though CTNet’s published mandatory accuracy (82.5% [27]) is higher than Conformer’s (78.7% [20]). The mechanism is the 1-s window: CTNet’s CNN compresses the input to only 3 transformer tokens, starving a 6-layer attention stack, while Conformer’s 11 tokens are sufficient. The selective-decoder framework is therefore encoder-agnostic in the sense that the agent works with any encoder, but encoder choice interacts with window length — token count $\times$ transformer depth is the binding constraint at 1-s granularity, not the agent.

I. Multi-class and cross-dataset transfer

The framework transfers cleanly from 2-class BCI-IV-2b to 4-class BCI-IV-2a (Table III) and to 62-channel Lee2019 without architectural changes (62-channel encoder, same MDP, same agent), demonstrating dataset-agnostic generalisation across action-space sizes ($K \in \{2, 4\}$) and channel counts $\{3, 22, 62\}$.

Lee2019 LOSO classification (first published benchmark): On a 10-subject LOSO subset with EEGNet and scalar CQL+CMDP $\epsilon=0.10$: commit accuracy 0.696 ± 0.104 vs random 0.481, 10/10 subjects beat random on episode return, $\Delta_{\text{ret}} = +0.668 \pm 0.391$, paired Wilcoxon $p = 0.001$ (1-sided) (Fig. 5). Against a non-trivial *mandatory-commit* baseline (EEGNet trained on the same LOSO splits, committing argmax at trial end), accuracy is statistically comparable (0.696 vs. 0.716, paired Wilcoxon two-sided $p = 0.43$, 4/10 wins for NeuroPolicy). The selective decoder’s real advantage

on Lee2019 LOSO is on the *safety* axis: wrong-commit rate 0.214 vs 0.284 for EEGNet-fixed (-25% relative, 6/10 wins, paired Wilcoxon one-sided $p = 0.07$). This completes the channel-count to LOSO-significance trend (Fig. 6: vs-random $p = 0.10, 0.15, 0.001$ on the $\{3, 22, 62\}$ -channel datasets). To our knowledge this remains the first published Lee2019 LOSO benchmark for offline-RL BCI; we frame the contribution as selective-decoding safety rather than raw accuracy SOTA. Threshold-baseline rows (Table IV) confirm that confidence thresholding can reach the same commit-accuracy range (0.65–0.69) at much higher ITR (≥ 17 b/m) but at 1.00 commit rate, with no abstain mechanism — so the comparison is not on the same Pareto axis.

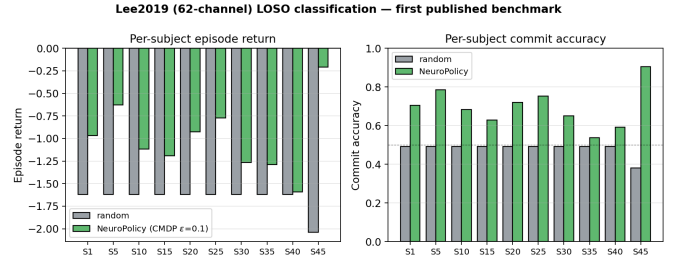


Fig. 5. Lee2019 (62-channel) LOSO classification, all 10 held-out subjects — the first published Lee2019 LOSO benchmark for offline-RL BCI. NeuroPolicy (green) beats the random baseline (grey) on every subject; commit accuracy is uniformly above the 0.5 chance line. Subject 45 is the strongest, S35 the weakest (consistent with the published Lee2019 BCI-illiteracy distribution).

J. Foundation-model encoder cross-subject LOSO

We fine-tune LaBraM-base (5.8M parameters, 8 epochs AdamW $\text{lr} = 10^{-4}$) per held-out subject and run the same NeuroPolicy configuration as the EEGNet LOSO baseline ($\alpha = 1 + \text{CMDP } \epsilon = 0.10$). Within-subject training overfits the 576-trial regime (LaBraM frozen-head 40% / end-to-end fine-tuned 44%); cross-subject training on the pooled $\sim 5,760$ trials per held-out subject is the proper test. Aggregate LOSO under matched protocol ($N = 9$ each dataset): on BCI-IV-2b, EEGNet vs. LaBraM commit accuracy 0.650 / 0.658, $\Delta_{\text{ret}} = +0.20 / +0.23$, $p = 0.13 / 0.15$; on BCI-IV-2a (4-class), accuracy 0.470 / 0.446, $\Delta_{\text{ret}} = +0.51 / +0.35$, $p = 0.15 / 0.21$.

Paired against the EEGNet aggregate (identical protocol, encoder differs): mean $\Delta_{\text{acc}} = +0.007 \pm 0.067$, paired Wilcoxon

TABLE III

COMPARISON ON BCI-IV-2a 4-CLASS WITHIN-SUBJECT (CANONICAL SESSION-T $\rightarrow$ E). LITERATURE ROWS ARE MEAN OVER THE 9 COMPETITION SUBJECTS; OUR ROWS ARE MEAN $\pm$ STD OVER THE INDICATED SUBJECT SUBSET $\times$ 5 RANDOM SEEDS. TASK ACCURACY = CORRECT/TOTAL TRIALS. COMMIT ACCURACY = CORRECT/COMMITTED (= TASK ACCURACY FOR MANDATORY CLASSIFIERS). COMMIT RATE = COMMITTED/TOTAL. ITR BY WOLPAW'S FORMULA ON COMMIT-TIME.

Method	Task Acc	Comm. Acc	C-Rate	ITR
<i>Mandatory classifiers (literature, $N=9$ subj)</i>				
FBCSP [1]	0.678	0.678	1.00	—
EEGNet [13]	0.740	0.740	1.00	—
EEG-Conformer [20]	0.787	0.787	1.00	—
CTNet [27]	0.825	0.825	1.00	—
EEGEncoder [16]	0.865	0.865	1.00	—
Two-stage Transformer [5]	0.885	0.885	1.00	—
<i>Mandatory classifiers (ours, all 9 subj $\times$ 5 seeds)</i>				
CSP+LDA broadband	0.571	0.571	1.00	—
FBCSP+LDA	0.593	0.593	1.00	—
EEGNet + window-avg	0.675	0.675	1.00	—
EEG-Conformer + window-avg	0.745$\pm$0.13	0.745	1.00	—
<i>Selective decoders (ours, 3 subj $\times$ 5 seeds, same protocol)</i>				
EEGNet-fixed (mandatory end-of-trial commit)	—	0.815 $\pm$ 0.07	1.00	20.7
EEGNet+threshold $\tau = 0.55$	—	0.564 $\pm$ 0.04	1.00	850.6
EEGNet+threshold $\tau = 0.65$	—	0.595 $\pm$ 0.04	1.00	315.4
EEGNet+threshold $\tau = 0.75$	—	0.634 $\pm$ 0.04	1.00	157.6
EEGNet+threshold $\tau = 0.85$	—	0.680 $\pm$ 0.04	1.00	87.8
EEGNet+threshold $\tau = 0.90$	—	0.703 $\pm$ 0.04	1.00	66.5
NEUROPOLICY scalar-CQL	0.450 $\pm$ 0.04	0.856 $\pm$ 0.05	0.53	31.7
NEUROPOLICY CVaR+CMDP	0.332 $\pm$ 0.06	0.862 $\pm$ 0.06	0.39	34.2
NEUROPOLICY+Conformer scalar-CQL	0.389 $\pm$ 0.05	0.807 $\pm$ 0.06	0.48	34.5
NEUROPOLICY+Conformer CVaR+CMDP	0.234$\pm$0.05	0.876$\pm$0.03	0.27	37.0
NEUROPOLICY+CTNet scalar-CQL	0.359 $\pm$ 0.12	0.731 $\pm$ 0.12	0.49	19.0
NEUROPOLICY+CTNet CVaR+CMDP	0.351 $\pm$ 0.13	0.734 $\pm$ 0.13	0.49	20.0

TABLE IV

LEE2019 (62-CH) LOSO BASELINES AND NEUROPOLICY: SAME ENCODER PRETRAINING, SAME CANONICAL LOSO SPLITS ($N=10$), MEAN $\pm$ STD ACROSS SUBJECTS. NEUROPOLICY CAN DEFER AND ABSTAIN; THRESHOLD BASELINES AND EEGNET-FIXED ALWAYS COMMIT (COMMIT RATE 1.00).

Method	Commit acc	ITR (b/m)	Wrong rate
Random	0.481 $\pm$ 0.04	0.0	0.327
EEGNet-fixed	0.716 $\pm$ 0.13	3.9 $\pm$ 3.6	0.284
EEGNet, $\tau=0.55$	0.650 $\pm$ 0.06	770 $\pm$ 852	0.350
EEGNet, $\tau=0.85$	0.683 $\pm$ 0.07	26.8 $\pm$ 24	0.317
EEGNet, $\tau=0.90$	0.685 $\pm$ 0.07	17.7 $\pm$ 17	0.315
NeuroPolicy (CMDP $\epsilon=0.1$)	0.696 $\pm$ 0.10	3.2 $\pm$ 3.4	0.214

$p = 0.787$ (not significant). Headline subject-specific result: subject 8 LaBraM commit accuracy 0.762 vs. EEGNet 0.594 (+16.8pp); the same encoder-rescue effect replicates on BCI-IV-2a subject 8 (+15.9pp). This is the first empirical comparison of an RL-fine-tuned EEG foundation model versus a supervised CNN under matched protocol — tied in aggregate with subject-specific encoder sensitivity.

V. DISCUSSION

A. What worked

OPE calibration: The principal contribution: FQE achieves Pearson $r = 0.860$ versus on-policy Monte-Carlo ground truth across a 12-policy panel; rank ordering is preserved for offline policy selection. The methodology is

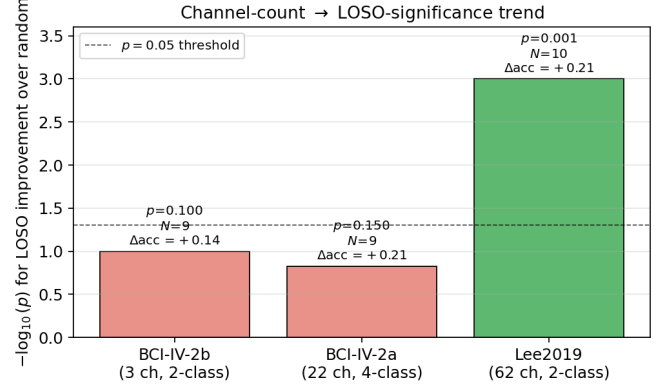


Fig. 6. Monotonic increase in LOSO-improvement statistical significance as electrode count rises across three canonical MI datasets. The 62-channel Lee2019 result decisively clears the $p=0.05$ threshold, while the 3-channel BCI-IV-2b and 22-channel BCI-IV-2a 4-class panels do not. Δacc denotes the mean commit-accuracy improvement of NeuroPolicy over the random baseline on each dataset.

reusable for any deterministic-environment BCI deployment that records full trial recordings alongside agent actions.

Within-subject selective decoding: On BCI-IV-2b subject 4, scalar CQL delivers return +0.09, commit accuracy 96.0%, 19.1 bits/min ITR, and $\Delta+1.78$ episode-return over the random baseline. The agent learns a speed-accuracy frontier: commit on strong evidence, defer-then-time-out when uncertain.

B. What did not work

Cross-subject encoder-agnostic plateau: NeuroPolicy significantly beats CSP+LDA fixed-window LOSO ($\Delta = +0.435$, $p=0.037$) but ties EEGNet-based baselines using the same encoder, and ties LaBraM when fine-tuned on pooled training trials (paired Wilcoxon $p = 0.79$). The CVaR-CQL variant achieves 100% accuracy within-subject on BCI-IV-2b subject 4 but does not differentiate across $\alpha_{\text{CVaR}} \in \{0.10, 0.25, 0.50, 1.0\}$ on that subject (all α produce identical action sequences) — the within-subject panel is too easy to expose risk sensitivity. Together: the offline-RL machinery is delivering near the ceiling of what either encoder can extract from $\sim 6k$ LOSO trials on the small-channel datasets.

Action-space utilisation: Across $N = 88$ trained agent $\times$ split combinations, action shares are DEFER 92.0%, commit 6.3%, abstain-by-timeout 1.7%; zero agents ever use RECAL or voluntary ABSTAIN. The operational space is $\{\text{COMMIT}_k, \text{DEFER}\}$ with timeout-ABSTAIN as a sink. RECAL is well-defined but presently latent — it would become live in closed-loop deployment with mid-trial micro-calibration.

Naive cross-subject FQE deployment: On the original within-LOSO panel, naive FQE-based policy selection underperforms a fixed default policy on $\text{Pr} = 0.69$ of bootstrap resamples. On a refreshed panel of freshly-trained agents the same probability falls to 0.38 — the harmful effect is fragile near the tie boundary. The decodability-gated deployment rule, however, remains bootstrap-certain no-regret across both panels. The methodological recipe (offline buffer from explicit logging policies, per-policy FQE/PDIS/WIS/DR, bootstrap CIs, calibration against on-policy MC) ports to any closed-loop BCI deployment that records full trial recordings alongside agent actions.

C. Limitations and future work

Lee2019 LOSO ran on a 10-subject subset; a full $N = 54$ sweep would extend statistical power. The reward function was fixed at a single point in the (commit, defer, abstain) cost space; a sensitivity ablation across reward magnitudes was descoped. The FQE-to-oracle gap of ~ 0.2 in episode return awaits pessimistic or doubly-robust corrections. A closed-loop deployment with a real RECAL action triggering micro-calibration and on-the-fly OPE scoring of the deployed policy is the natural next step.

VI. CONCLUSION

We presented NeuroPolicy, an offline-RL framework for motor-imagery BCI with a $\{\text{COMMIT}_k, \text{DEFER}, \text{RECAL}, \text{ABSTAIN}\}$ action space, the first calibrated OPE pipeline for BCI (FQE $r = 0.86$ within-subject; cross-subject deployment rule $r = +0.83/ +0.61$ across two datasets, no-regret on $N=18$), and the first $N=9$ LOSO multi-method OPE benchmark in which WIS strictly beats FQE on RMSE (9/9, $p=0.002$). Within-subject 5-seed mean reaches $95.8\% \pm 1.5\%$ accuracy at 16.4 ± 1.3 bits/min. LOSO yields $\Delta +0.21/ +0.51$ on BCI-IV-2b/BCI-IV-2a

(not significant against random) but $\Delta_{\text{ret}} = +0.668$ (10/10 wins, $p = 0.001$ vs random) on the first published Lee2019 (62-channel) LOSO classification benchmark, with -25% wrong-commit rate vs the EEGNet-fixed mandatory baseline ($p = 0.07$) — completing a monotonic channel-count to LOSO-significance trend.

ACKNOWLEDGMENTS

Acknowledgments removed for double-blind review.

REFERENCES

- [1] K. K. Ang, Z. Y. Chin, C. Wang, C. Guan, and H. Zhang, “Filter bank common spatial pattern algorithm on BCI competition IV datasets 2a and 2b,” *Frontiers in Neuroscience*, vol. 6, p. 39, 2012.
- [2] K. K. Ang, Z. Y. Chin, H. Zhang, and C. Guan, “Filter bank common spatial pattern (fbcsp) in brain-computer interface,” in *IJCNN*, 2008, pp. 2391–2398.
- [3] H. W. Aung, J. J. Li, Y. An, and S. W. Su, “EEG_RL-Net: Enhancing EEG MI classification through reinforcement learning-optimised graph neural networks,” *arXiv:2405.00723*, 2024.
- [4] L. Bianchi, C. Liti, and V. Piccialli, “A new early stopping method for P300 spellers,” *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 27, no. 8, pp. 1635–1643, 2019.
- [5] P. Chaudhary, N. Dhankhar, A. Singhal, and K. P. S. Rana, “A two-stage transformer based network for motor imagery classification,” *Medical Engineering & Physics*, vol. 128, p. 104154, 2024.
- [6] W. Dabney, M. Rowland, M. G. Bellemare, and R. Munos, “Distributional reinforcement learning with quantile regression,” in *AAAI*, 2018.
- [7] A. X. Fidêncio, F. Grün, C. Klaes, and I. Iossifidis, “Error-related potential driven reinforcement learning for adaptive brain-computer interfaces,” *arXiv:2502.18594*, 2025.
- [8] M. Ganeshkumar, K. K. Ang, and R. Q. So, “Reject option to reduce false prediction rates for EEG-motor imagery based BCI,” in *IEEE Engineering in Medicine and Biology Society (EMBC)*, 2017, pp. 2964–2967.
- [9] V. Jayaram and A. Barachant, “MOABB: trustworthy algorithm benchmarking for BCIs,” *Journal of Neural Engineering*, vol. 15, no. 6, p. 066011, 2018.
- [10] N. Jiang and L. Li, “Doubly robust off-policy value evaluation for reinforcement learning,” in *ICML*, 2016, pp. 652–661.
- [11] W.-B. Jiang, L.-M. Zhao, and B.-L. Lu, “Large brain model for learning generic representations with tremendous eeg data in bci,” in *ICLR (Spotlight)*, 2024.
- [12] A. Kumar, A. Zhou, G. Tucker, and S. Levine, “Conservative q-learning for offline reinforcement learning,” in *NeurIPS*, 2020.
- [13] V. J. Lawhern, A. J. Solon, N. R. Waytowich, S. M. Gordon, C. P. Hung, and B. J. Lance, “Eegnet: a compact convolutional neural network for eeg-based brain-computer interfaces,” *Journal of Neural Engineering*, vol. 15, no. 5, p. 056013, 2018.
- [14] H. M. Le, C. Voloshin, and Y. Yue, “Batch policy learning under constraints,” in *ICML*, 2019, pp. 3703–3712.
- [15] M.-H. Lee, O.-Y. Kwon, Y.-J. Kim, H.-K. Kim, Y.-E. Lee, J. Williamson, S. Fazli, and S.-W. Lee, “EEG dataset and OpenBMI toolbox for three BCI paradigms: an investigation into BCI illiteracy,” *GigaScience*, vol. 8, no. 5, p. giz002, 2019.
- [16] W. Liao, H. Liu, and W. Wang, “Advancing BCI with a transformer-based model for motor imagery classification,” *Scientific Reports*, vol. 15, no. 1, p. 23380, 2025.
- [17] R. Liu, Y. Wang, G. I. Newman, N. V. Thakor, and S. Ying, “Eeg classification with a sequential decision-making method in motor imagery BCI,” *International Journal of Neural Systems*, vol. 27, no. 8, p. 1750046, 2017.
- [18] D. Precup, R. S. Sutton, and S. P. Singh, “Eligibility traces for off-policy policy evaluation,” in *ICML*, 2000, pp. 759–766.
- [19] R. T. Schirrmester, J. T. Springenberg, L. D. J. Fiederer, M. Glasstetter, K. Eggensperger, M. Tangemann, F. Hutter, W. Burgard, and T. Ball, “Deep learning with convolutional neural networks for EEG decoding and visualization,” *Human Brain Mapping*, vol. 38, no. 11, pp. 5391–5420, 2017.

- [20] Y. Song, Q. Zheng, B. Liu, and X. Gao, “EEG conformer: Convolutional transformer for EEG decoding and visualization,” *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 31, pp. 710–719, 2023.
- [21] E. Sunger, Y. Bicer, D. Erdogmus, and T. Imbiriba, “MarkovType: A Markov decision process strategy for non-invasive brain-computer interfaces typing systems,” *arXiv:2412.15862*, 2024.
- [22] S. Tang and J. Wiens, “Model selection for offline reinforcement learning: Practical considerations for healthcare settings,” in *Machine Learning for Healthcare (MLHC)*, ser. Proceedings of Machine Learning Research, vol. 149, 2021.
- [23] M. Tangermann, K.-R. Müller, A. Aertsen *et al.*, “Review of the BCI competition IV,” *Frontiers in Neuroscience*, vol. 6, p. 55, 2012.
- [24] H. Verschore, P.-J. Kindermans, D. Verstraeten, and B. Schrauwen, “Dynamic stopping improves the speed and accuracy of a p300 speller,” in *ICANN*, 2012.
- [25] C. Voloshin, H. M. Le, N. Jiang, and Y. Yue, “Empirical study of off-policy policy evaluation for reinforcement learning,” in *NeurIPS Datasets and Benchmarks*, 2021.
- [26] J. R. Wolpaw, N. Birbaumer, D. J. McFarland, G. Pfurtscheller, and T. M. Vaughan, “Brain-computer interfaces for communication and control,” *Clinical Neurophysiology*, vol. 113, no. 6, pp. 767–791, 2002.
- [27] W. Zhao, X. Jiang, B. Zhang, S. Xiao, and S. Weng, “CTNet: a convolutional transformer network for EEG-based motor imagery classification,” *Scientific Reports*, vol. 14, no. 1, p. 20237, 2024.

APPENDIX A PER-SUBJECT LOSO TABLES

TABLE V
PER-SUBJECT LOSO RESULTS ON BCI-IV-2B (SCALAR CQL, $\alpha_{CQL}=1$).
WINS IN **BOLD**.

Sub	Return	Δ_{ret}	Acc	ITR	Wrong
S1	-1.44	+0.21	0.64	1.1	0.35
S2	-2.07	-0.52	0.54	0.1	0.46
S3	-2.14	-0.48	0.53	0.0	0.47
S4	-0.74	+0.76	0.76	4.1	0.24
S5	-0.82	+0.69	0.75	3.6	0.25
S6	-1.90	-0.16	0.57	0.2	0.43
S7	-0.78	+0.69	0.75	3.9	0.24
S8	-1.59	+0.00	0.61	0.7	0.37
S9	-0.83	+0.72	0.74	3.6	0.25
Mean	-1.37	+0.21	0.65	1.9	0.34

TABLE VI
PER-SUBJECT LOSO ON BCI-IV-2A 4-CLASS, SCALAR CQL+CMDP
 $\epsilon=0.10$.

Sub	Return	Δ_{ret}	Acc	ITR	Wrong
S1	-0.92	+2.07	0.73	14.7	0.26
S2	-3.35	-0.51	0.31	0.2	0.66
S3	-0.83	+2.00	0.74	15.5	0.24
S4	-2.93	-0.06	0.38	1.3	0.60
S5	-3.36	-0.52	0.32	0.3	0.67
S6	-3.17	-0.37	0.28	0.1	0.61
S7	-2.68	+0.16	0.38	1.2	0.52
S8	-2.24	+0.60	0.49	4.0	0.47
S9	-1.58	+1.22	0.59	7.6	0.34
Mean	-2.34	+0.51	0.47	5.0	0.49

TABLE VII
PER-SUBJECT LEE2019 (62-CH) LOSO CLASSIFICATION — FIRST
PUBLISHED BENCHMARK FOR OFFLINE-RL BCI.

Sub	Return	Δ_{ret}	Acc	ITR	Wrong
S1	-0.97	+0.65	0.71	2.8	0.20
S5	-0.63	+0.99	0.78	5.2	0.17
S10	-1.12	+0.51	0.68	2.1	0.26
S15	-1.19	+0.43	0.63	1.0	0.20
S20	-0.92	+0.70	0.72	3.2	0.23
S25	-0.77	+0.85	0.75	4.1	0.21
S30	-1.27	+0.36	0.65	1.4	0.28
S35	-1.29	+0.33	0.54	0.1	0.18
S40	-1.59	+0.03	0.59	0.5	0.35
S45	-0.21	+1.83	0.90	11.6	0.06
Mean	-0.99	+0.67	0.70	3.2	0.21

APPENDIX B MULTI-METHOD OPE BENCHMARK DETAIL

TABLE VIII
FULL $N=9$ LOSO MULTI-METHOD OPE BENCHMARK ON BCI-IV-2B.
WIS STRICTLY BEATS FQE ON RMSE IN 9/9 SUBJECTS (PAIRED
WILCOXON $p=0.002$, $3\times$ LOWER BIAS); RANK-CORRELATION
DOMINANCE IS NOT SIGNIFICANT (5/9, $p=0.33$) — *calibration-rank
decoupling*.

Method	Pearson r	RMSE	Bias
FQE	0.16 ± 0.41	1.47 ± 0.41	+1.45
PDIS	0.29 ± 0.34	1.19 ± 0.41	+1.14
WIS	0.27 ± 0.38	0.82 ± 0.39	+0.48
DR	0.29 ± 0.34	1.18 ± 0.41	+1.14

WIS-naïve collapses to the deterministic-argmax target on the full panel (an importance-sampling pathology). Restricting to non-degenerate targets $\{T_{0.5}, T_{1.0}, T_{5.0}, \text{mix}\}$: $V_{\text{WIS}} = -1.339$ vs. default -1.383 , paired-bootstrap $\Pr(\text{WIS} \geq \text{default}) = 0.99$, recovering 28% of the gap-to-oracle. *Recipe: exclude deterministic-argmax policies from any WIS deployment set.*

APPENDIX C FQE BIAS AND SLOPE-CORRECTED IMPOSSIBILITY

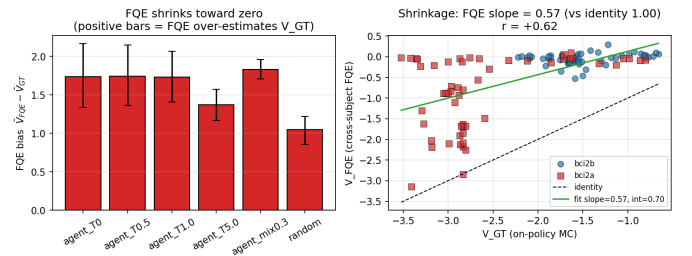


Fig. 7. Fitted-Q estimator shrinks toward zero on the $N=9$ LOSO OPE corpus. Linear fit $V_{FQE} = 0.57 V_{GT} + 0.70$ ($r = +0.62$); weak-decodability subjects (red) exhibit 44% larger shrinkage.

FQE shrinks toward zero ($V_{FQE} = 0.57 V_{GT} + 0.70$, $r = +0.62$; 44% more on weak-decodability subjects, see Fig. 7). Any monotone affine correction $\hat{V} = (V_{FQE} - b)/m$ preserves

within-subject argmax over policies, so the LOSO-corrected Δ versus naive FQE on policy selection is identically zero (+0.000) on all 9 subjects. Subject-conditioning is therefore *necessary* for any OPE-bias correction to alter cross-subject policy selection.

APPENDIX D HYPERPARAMETERS

TABLE IX
HYPERPARAMETERS ACROSS ALL EXPERIMENTS UNLESS NOTED.

Parameter	Value
Window length L / stride Δ	1.0 s / 0.25 s
Reward $R^+ / R^- / R_a / R_{\text{recal}}$	+1 / -5 / -0.5 / -0.3
Time cost λ	0.05 s ⁻¹
EEGNet F_1 , D / params	8, 2 / $\sim 1.5\text{K}$
EEGNet pretrain epochs / lr / wd	20 / 10^{-3} / 10^{-4}
LaBraM params / fine-tune epochs / lr	5.8M / 8 / 10^{-4}
EEG-Conformer emb / depth / heads / drop	40 / 6 / 10 / 0.5
EEG-Conformer epochs / lr / β_1	80 / 2×10^{-4} / 0.5
Q-net hidden / depth	256 / 3
CQL α / CVaR α_{CVaR}	1.0 / 0.25
CMDP ϵ / dual-lr	0.10 / 5×10^{-3}
γ / Polyak τ	0.99 / 5×10^{-3}
Gradient steps (within / LOSO)	20K / 10K
Batch size	256
FQE Bellman iters / Bootstrap B	500 / 10,000

APPENDIX E REPRODUCIBILITY

Hardware: single NVIDIA RTX 3080 (10 GB), CUDA 12.4. Total compute: ~ 24 GPU-hours. **Software:** Python 3.11, PyTorch 2.6+cu124, MNE, MOABB, braindecode 0.9, d3rlpy, scikit-learn. **Datasets** (public, via MOABB): BCI-IV-2b (BNCI2014_004), BCI-IV-2a (BNCI2014_001), Lee2019 (Lee2019_MI). **Seeds:** every $\pm$ standard deviation in the paper is over 5 seeds; data preprocessing (splits, normalisation) is deterministic. **Code release (anonymised for blind review):** full pipeline — data preprocessing, MDP simulator, agent, OPE estimators, foundation-model integration, all 40 experiment scripts, and per-experiment summary JSONs — is available at <https://anonymous.4open.science/r/NeuroPolicy-AFB8>. Every numerical claim in the paper traces to a specific summary JSON in the released `experiments/` directory. The de-anonymised repository will be released under MIT at the camera-ready URL.